

Smart AI Medical Report

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Patient Profile

Saubhik Bhaumik, 27-year-old male

Summary of Results

Liver function tests demonstrate values entirely within reference ranges, indicating preserved hepatic function. The predictive model assesses the profile as within normal limits.

Detailed Test Explanations

SERUM BILIRUBIN (TOTAL): Measures overall bilirubin levels, a byproduct of red blood cell breakdown processed by the liver; elevations suggest hepatocellular damage, biliary obstruction, or hemolysis, while low levels are uncommon and typically insignificant. Patient result: 0.9 mg/dL (reference: 0.2-1.2 mg/dL), normal.

SERUM BILIRUBIN (DIRECT): Quantifies conjugated bilirubin excreted in bile; increased levels indicate biliary obstruction or intrahepatic cholestasis. Patient result: 0.2 mg/dL (reference: 0-0.3 mg/dL), normal.

SERUM BILIRUBIN (INDIRECT): Reflects unconjugated bilirubin from hemolysis or impaired hepatic uptake/conjugation; elevations point to hemolytic anemia or Gilbert syndrome. Patient result: 0.70 mg/dL (reference: 0.2-1.0 mg/dL), normal.

SGPT (ALT): Alanine aminotransferase, a liver-specific enzyme; elevations signify hepatocellular injury from viral hepatitis, toxins, or ischemia. Patient result: 36 U/L (reference: 13-40 U/L), normal.

SGOT (AST): Aspartate aminotransferase, present in liver, heart, and muscle; rises indicate tissue damage, though less specific to liver than ALT. Patient result: 32 U/L (reference: 0-37 U/L), normal.

SERUM ALKALINE PHOSPHATASE: Enzyme elevated in cholestasis, bone disease, or pregnancy; not assessed in this panel.

SERUM PROTEIN: Total protein reflects synthetic liver function and nutritional status; hypoalbuminemia or hypoproteinemia signals chronic liver disease or malnutrition. Patient result: 7.2 g/dL (reference: 6.4-8.3 g/dL), normal.

SERUM ALBUMIN: Primary protein synthesized by hepatocytes, indicator of chronic liver synthetic capacity and nutritional state; decreases in cirrhosis or nephrotic syndrome. Patient result: 4.7 g/dL (reference: 3.5-5.2 g/dL), normal.

GLOBULIN: Immunoglobulin fraction; alterations occur in chronic inflammation, infections, or liver disease. Patient result: 2.50 g/dL (reference: 1.8-3.6 g/dL), normal.

A/G RATIO: Albumin-to-globulin ratio; imbalance suggests dysproteinemia or chronic disease. Patient result: 1.88 (reference: 1.1-2.4), normal.

Overall Interpretation

Comprehensive liver function panel yields normal results across bilirubin fractions, transaminases, proteins, and albumin-globulin balance, consistent with intact hepatic integrity and synthetic function. The predictive model assesses the profile as within normal limits, supporting absence of overt liver pathology; clinical correlation advised per laboratory remarks.

Recommendations

Maintain balanced diet rich in fruits, vegetables, and lean proteins; limit alcohol intake to moderate levels; engage in regular aerobic exercise (150 minutes weekly); monitor weight and avoid hepatotoxic medications without medical supervision. Routine annual hepatic screening recommended; consult physician for any symptoms such as fatigue, jaundice, or abdominal pain.

Disclaimer

This report provides interpretive analysis based on provided data and is not a substitute for professional medical evaluation, diagnosis, or treatment.